Summary

Memorandum on Northline Cache for repeated lookups.

Analysts were waiting on repeated semantic lookups against a fixed desk corpus.

Northline Cache reduced p95 lookup latency from 180 ms to 74 ms by retaining embeddings for repeated queries.

The March replay showed a 2.1 point gain in recall at five after the cache warmed.

Operator decision: Keep the warm store enabled for the analyst desk and bypass it for one-off research queries..

Evidence

Method note: Latency was measured on 400 repeated queries against a frozen 12,000-chunk corpus, with a cold run before each cached run.

Reported result: Median latency fell from 96 ms to 41 ms and recall at five rose from 0.71 to 0.73.

Limits of the evidence: The cache measurement used one corpus snapshot and did not cover streaming updates.

Open questions: Does the latency gain hold when more than one analyst shares the same warm store? Risk review: Severity medium: A stale cached vector can surface a superseded paragraph after a document is replaced.

Recommended action: Invalidate cache entries whenever a source file hash changes..